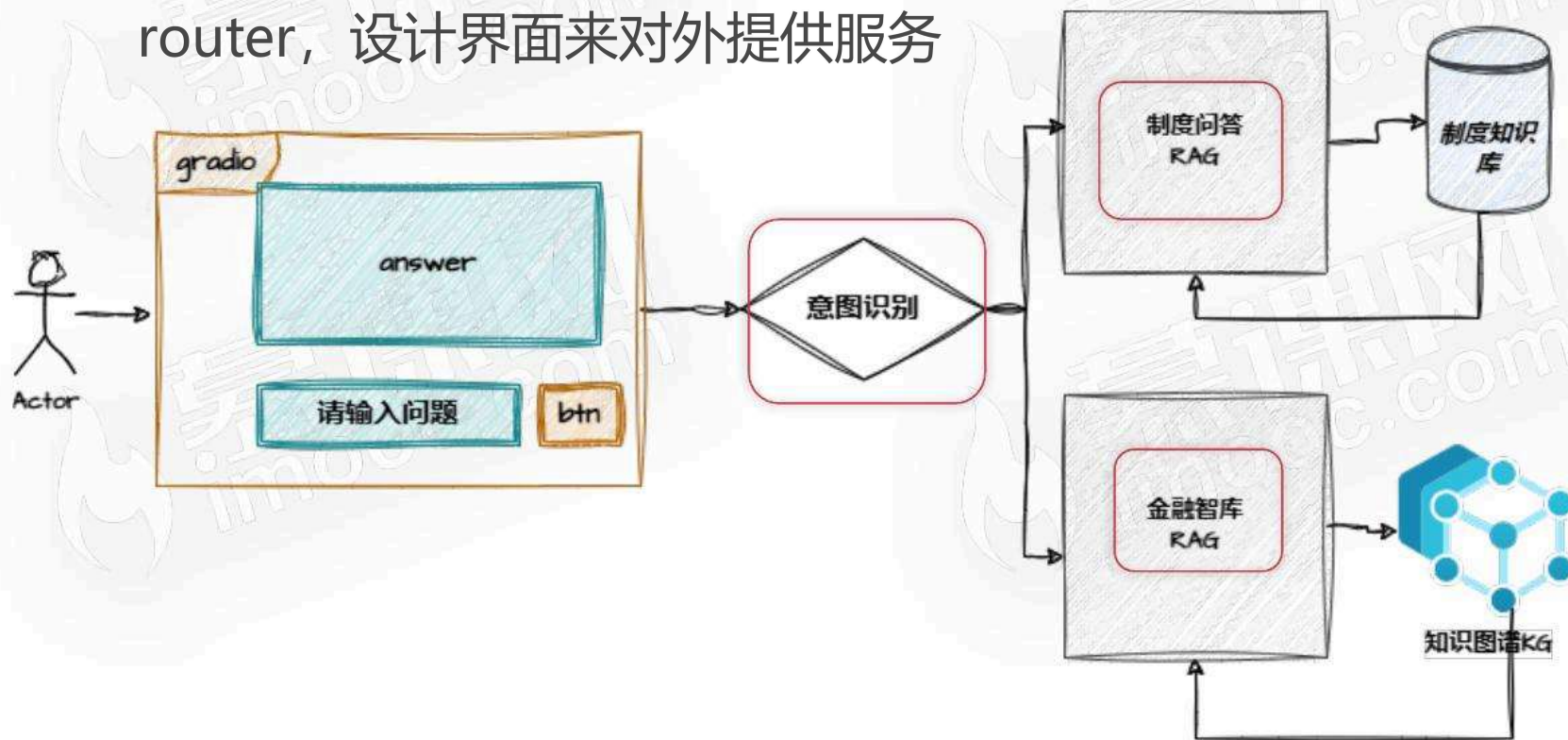


企业员工助手 界面开发

本章简介

- ◆ 整合制度问答组手和基于知识图谱的金融智库，以及RAG-router，设计界面来对外提供服务



本章简介

◆ Gradio介绍

◆ 实战



◆ 界面设计

◆ 部署

Gradio介绍

◆ gradio

gradio



一个强大的Python库，专注于快速创建简单、直观的用户web界面，让用户可以方便地与机器学习模型进行交互



Gradio介绍

◆ gradio

```
import gradio as gr

# 定义处理函数
def greet(name, checked):
    if checked:
        return f"Hello, {name}! You have checked the box."
    else:
        return f"Hello, {name}! You have not checked the box."

# 创建Gradio应用
with gr.Blocks() as demo:
    # 输入组件
    name_input = gr.Textbox(label="Name") # 文本框, 用于输入名字
    checkbox = gr.Checkbox(label="Check me") # 复选框, 用于选择是否勾选

    # 输出组件
    output_text = gr.Textbox(label="Output Box") # 文本框, 用于显示输出结果

    # 按钮组件及事件绑定
    greet_btn = gr.Button("Greet") # 创建按钮
    greet_btn.click(fn=greet, # 绑定按钮点击事件, 设置输入和输出组件
                    inputs=[name_input, checkbox],
                    outputs=[output_text])

# 启动Gradio应用
if __name__ == "__main__":
    demo.launch()
```



Gradio介绍

◆ Gradio-事件处理

```
name_input = gr.Textbox(label="Name") # 文本框, 用于输入名字
checkbox = gr.Checkbox(label="Check me") # 复选框, 用于选择是否勾选

# 输出组件
output_text = gr.Textbox(label="Output Box") # 文本框, 用于显示输出结果

# 按钮组件及事件绑定
greet_btn = gr.Button("Greet") # 创建按钮
greet_btn.click(fn=greet, inputs=[name_input, checkbox], outputs=[output_text])

def greet(name, checked):
    if checked:
        return f"Hello, {name}! You have checked the box."
    else:
        return f"Hello, {name}! You have not checked the box."
```

输入组件: 文本框和
单选框

传入处理函
数的是对应
组件的值

Gradio介绍

◆ Gradio-布局-Row and Column

```
import gradio as gr
with gr.Blocks() as demo:
```

1行

```
    with gr.Row():
```

```
        with gr.Column(scale=1):
```

```
            text1 = gr.Textbox()
```

```
            text2 = gr.Textbox()
```

```
        with gr.Column(scale=4):
```

```
            btn1 = gr.Button("Button 1")
```

```
            btn2 = gr.Button("Button 2")
```

2列, 2列比例
scale=1:4

```
if __name__ == "__main__":
    demo.launch()
```

The image shows a Gradio interface with two columns. The left column contains two text boxes, each labeled 'Textbox'. The right column contains two buttons, labeled 'Button 1' and 'Button 2'. The buttons are wider than the text boxes, illustrating the scale=1:4 ratio mentioned in the code.

Gradio介绍

◆ Gradio-chatbot组件

```
import gradio as gr
```

```
def load():
```

```
    return [
```

```
        ("Here's an audio", "audio"),
```

```
        ("Here's an video", "video")
```

```
    ]
```

```
with gr.Blocks() as demo:
```

```
    chatbot = gr.Chatbot()
```

```
    button = gr.Button("Load audio and video")
```

```
    button.click(load, None, chatbot)
```

```
demo.launch()
```

(query, answer)

Chatbot

Here's an audio

audio

Here's an video

video

界面设计

◆ 界面设计

上下文信息

chatbot



界面设计

◆ 界面设计

上下文信息

chatbot



项目部署

◆ 部署

uvicorn



基于asyncio库开发的轻量级且高效的
ASGI (Asynchronous Server Gateway
Interface, 异步服务器网关接口) 服务器
, 用于构建异步Web服务

uvicorn



Uvicorn适用于各种类型的Web应用程序
, 特别是需要高性能异步请求处理的场景

项目部署

◆ 部署

step1

```
from fastapi import FastAPI
import gradio as gr
app = FastAPI()
@app.get("/")
def read_main():
    return {"message": "This is your main app"}
io = gr.Interface(lambda x: "Hello, " + x + "!", "textbox", "textbox")
app = gr.mount_gradio_app(app, io, path="/gradio")
```

erag.py: 将gradio
封装成fastapi 应用

step2

```
uvicorn erag:app --host 127.0.0.1 --port 7860 --workers 4
```

erag.py: 文件名

Ip和端口

进程数:
并发数

<http://127.0.0.1:7680/gradio>